

HANZE LI

GitHub: [Maotechh](#) | ShanghaiTech University

EDUCATION

ShanghaiTech University Sep 2024 - Present
B.S. Candidate | Computer Science Shanghai, China

METAL Lab (Prof. Junrui Liang) Sep 2025 - Present
Undergraduate Research Assistant | Self-powered systems / embedded TinyML Shanghai, China

Research & Engineering Profile: I research *self-powered embedded systems* and *TinyML*, and build complete embedded systems as a competition captain and sole developer, spanning *custom PCBs*, *LA32R out-of-order CPUs*, *FPGA/Linux*, and *vision control*, embedded firmware, and on-device inference.

HONORS

Best Paper Award - IEEE ICPS 2026 May 2026
Co-first author; battery-free IoT modeling / adaptive power path / task scheduling

National Student Computer System Capability Challenge (Loongson Cup) - Second Prize Aug 2026
Captain, GeekPie_; SpinalHDL / LA32R out-of-order CPU / 100 MHz FPGA / Linux

TI Cup Shanghai Undergraduate Electronic Design Competition - First Prize Aug 2026
Captain and vision/mechanical lead; K230 vision / mechanical design / STM32F407 integration

NVIDIA DGX Spark Hackathon - Excellence Award Aug 2026
System integration and local deployment; Qwen3.5 / Whisper / FastAPI / React / CUDA

NCTF 2025 Cybersecurity Challenge - 21st Place (Top 2% / 1046 Teams) Mar 2025
Reverse engineering and embedded security; x86 assembly / firmware analysis

39th Youth Science and Technology Innovation Contest - First Prize Jan 2024
Sole developer; embedded hardware / sensors / IoT environmental-monitoring prototype

15th VEX Robotics Asia Open Championship - First Prize Feb 2023
Core hardware/software/control member; C++ / PID / chassis and mechanisms

SKILLS

Expertise: Self-powered IoT systems | Energy harvesting | Embedded TinyML | Computer architecture | FPGA SoC design | Low-power design | Computer vision

Hardware: Schematic design | Multi-layer PCB layout (JLCEDA) | SMT & board bring-up | Power-path management | Signal conditioning | 3D CAD

Languages: C/C++ | Python | Scala/SpinalHDL | Verilog | Assembly (x86/RISC-V/LA32R)

Platforms: ESP32/STM32/nRF52/nRF54/K230 | Zephyr/TFLM/CanMV | FPGA/Vivado | Linux/Buildroot | BLE/Wi-Fi/MQTT | Git/Docker

RESEARCH HIGHLIGHTS

Battery-Free IoT Energy Management | CO-FIRST AUTHOR Best Paper @ IEEE ICPS 2026
Design framework and prototype optimization for battery-free IoT devices Aug 2025 - May 2026

- Proposed a two-dimensional (p , e) supply-demand state model for battery-free IoT systems and turned it into design rules for converter selection, control policy, and storage sizing.
- Built adaptive power-path and task-scheduling logic for PV-powered IoT devices, reducing system power consumption by 60% while preserving latency for critical tasks.
- Presented the resulting work at **IEEE ICPS 2026** as co-first author; awarded **Best Paper**.

PEAK Self-Powered TinyML Sensor Node SOLE DEVELOPER / RESEARCHER <i>Battery-free TinyML sensing platform; manuscript in preparation</i>	nRF54L15, TinyML <i>Dec 2025 - Present</i>
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- Independently developed the complete prototype across custom hardware, embedded firmware, TinyML inference, and BLE communication.
- Integrated energy-aware execution, on-device inference, and BLE reporting for intermittent embedded operation.

Anti-Poaching Risk Mapping SOLE DEVELOPER / RESEARCHER <i>Spatiotemporal patrol-priority prediction with Prof. Fei Fang (CMU) and WWF</i>	Carnegie Mellon x WWF <i>2026 - Present</i>
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- Independently developing a spatiotemporal decision-support system for wildlife-protection patrol prioritization in collaboration with Prof. Fei Fang at Carnegie Mellon University and WWF.

SYSTEMS & EMBEDDED HARDWARE PROJECTS

MIKU LA32R Out-of-Order SoC CAPTAIN / CPU & LINUX <i>SpinalHDL four-issue CPU and Linux system for the Loongson Cup final</i>	Second Prize · 100 MHz <i>Jul 2026 - Aug 2026</i>
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- Captained the four-member GeekPie__ team; optimized SpinalHDL LSQ/L1D, branch-prediction, and ROB timing for the LA32R SoC, achieving 100 MHz timing closure on Artix-7.
- Built a working FPGA Linux system with Linux 5.14, Buildroot, framebuffer/X11, PS/2, USB HID, and networking support.

SparkLive Local Multimodal Streaming System TEAM DEVELOPER / INTEGRATION <i>Live-stream assistance and digital-human platform running on DGX Spark</i>	Excellence Award · DGX Spark <i>Jul 2026 - Aug 2026</i>
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- Co-developed a DGX Spark platform integrating Qwen3.5, Whisper/VAD, OBS WebSocket, automatic clipping, TTS/Live2D, FastAPI/WebSocket, React, and PySide6; contributed multi-agent approval, fallback, and resumable CUDA/NVENC/NVDEC media processing.

H Problem: Ball-Balancing Vehicle CAPTAIN / VISION / MECHANICAL <i>Vision measurement and mechanical integration for a ball-balancing vehicle</i>	First Prize · K230 / F407 <i>Jul 2026 - Aug 2026</i>
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- Owned the CanMV K230 vision pipeline: implemented 640x480 grayscale ROI tracking, dark-blob filtering, beam centerline fitting, ball-center projection, and physical-coordinate calibration, reaching approximately 1 mm static error at 88.9 FPS.
- Integrated the grooved beam, hinge, screw-driven actuator, and CSI camera, and connected K230 measurement to STM32F407 closed-loop control.

PEAK Energy-Harvesting Power Management Board SOLE DEVELOPER <i>Custom power-management board for battery-free embedded operation</i>	172 nA · Custom PCB <i>Dec 2025 - Present</i>
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- Designed, assembled, and brought up the custom PCB; measured 172 nA quiescent current and integrated it with the battery-free sensing platform.

24-Channel Dual-Spectrum Line Sensor HARDWARE / FIRMWARE LEAD <i>High-density custom sensor board for high-speed robotic competitions</i>	PCB, SMT, MCU <i>Jun 2025</i>
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- Independently completed schematic design, PCB layout, SMT soldering, and firmware for the dense dual-spectrum sensor board.
- Implemented cascaded-latch I/O multiplexing for 24-channel quasi-synchronous sampling at 1 kHz, plus infrared / visible-light binarization and ambient-light filtering.

LEADERSHIP

GeekPie Association PRESIDENT <i>President of ShanghaiTech's student technology society; lead for embedded hardware initiatives</i>	Leadership <i>Sep 2025 - Present</i>
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- Lead association strategy, budgeting, external collaboration, and campus technology events; built the embedded hardware track around MCU development, PCB design, board bring-up, workshops, shared lab resources, and peer mentoring.